CANONICAL GAREL MAPS ON DIFFERENTIAL FORMS FOR QUASI-FINITE SYNTOMIC MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. The old dump stopped at3741; the exact cached chapter includes3742–3764 and the final universal-refinement step. The task uses the complete original range.

Editorial note. The closing diagram uses the already proved local-universal-family construction; it is an author-compressed descent step, not an inserted alternative proof.

Editorial note. The separate downstream trace map lemma-Garel is not counted here. The earlier Tate map is only a comparison of proof strategy; no free Garel map is imported.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Differentials, exterior powers and determinant lines; syntomic local complete-intersection presentations, etale locus, fibre-product algebra patching, and standard affine localization. The local funny-map construction, compatibility lemmas and universal family are retained author obligations.

Source reference. Original section 0FKB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19456–19456. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FK6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19451–19451. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FJI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19428–19428. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 00SP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1047–1047. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1053–1053. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1016–1016. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 07HX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9567–9567. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BWE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14942–14942. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1050–1050. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0FKD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19458–19458. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FK8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19453–19453. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FK9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19454–19454. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FKC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19457–19457. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3241–3241. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FKA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19455–19455. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0H9B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21521–21521. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10893–10893. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 069K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8044–8044. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01UF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2408–2408. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1054–1054. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00PK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 934–934. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1056–1056. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01UE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2407–2407. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FK3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19448–19448. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14907–14907. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 080Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10190–10190. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14908–14908. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BVG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14909–14909. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00OY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 912–912. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AX0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13738–13738. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A71: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12864–12864. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DW9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17389–17389. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DWA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17390–17390. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 960–960. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FJX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19442–19442. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FK0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19445–19445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FJY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19443–19443. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BVJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14912–14912. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 0BVB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14904–14904. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BT9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14832–14832. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *There exists a unique rule that to every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_Y$-module maps*

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for $p \geq 0$ satisfying (1), (2), and (3). In particular, the composition of $c^p_{Y/X}$ with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$.

1. The Tate map

In this section we produce an isomorphism between the determinant of the relative cotangent complex and the relative dualizing module for a locally quasi-finite syntomic morphism of locally Noetherian schemes. Following [1, 1.4.4] we dub the

isomorphism the Tate map. Our approach is to avoid doing local calculations as much as is possible.

Let $Y \to X$ be a locally quasi-finite syntomic morphism of schemes. We will use all the equivalent conditions for this notion given in Lemma 2.6 without further mention in this section. In particular, we see that $NL_{Y/X}$ is a perfect object of $D(\mathcal{O}_Y)$ with tor-amplitude in $[-1, 0]$. Thus we have a canonical invertible module $\det(NL_{Y/X})$ on Y and a global section

$$\delta(NL_{Y/X}) \in \Gamma(Y, \det(NL_{Y/X}))$$

See Derived Categories of Schemes, Lemma 0FJX. Suppose given a commutative diagram of schemes

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

whose vertical arrows are locally quasi-finite syntomic and which induces an isomorphism of Y' with an open of $X' \times_X Y$. Then the canonical map

$$Lb^* NL_{Y/X} \longrightarrow NL_{Y'/X'}$$

is a quasi-isomorphism by More on Morphisms, Lemma 0FK0. Thus we get a canonical isomorphism $b^* \det(NL_{Y/X}) \to \det(NL_{Y'/X'})$ which sends the canonical section $\delta(NL_{Y/X})$ to $\delta(NL_{Y'/X'})$, see Derived Categories of Schemes, Remark 0FJY.

Remark 1.1. Let $Y \to X$ be a locally quasi-finite syntomic morphism of schemes. What does the pair $(\det(NL_{Y/X}), \delta(NL_{Y/X}))$ look like locally? Choose affine opens $V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ and an integer n and $f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$ such that $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. Then

$$NL_{B/A} = \Big((f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} B\mathrm{d}x_i \Big)$$

and $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the classes $\overline{f}_i$. See proof of Lemma 2.6. Thus $\det(L_{B/A})$ is free on the generator

$$\mathrm{d}x_1 \wedge \ldots \wedge \mathrm{d}x_n \otimes (\overline{f}_1 \wedge \ldots \wedge \overline{f}_n)^{\otimes -1}$$

and the section $\delta(NL_{B/A})$ is the element

$$\delta(NL_{B/A}) = \det(\partial f_j/\partial x_i) \cdot \mathrm{d}x_1 \wedge \ldots \wedge \mathrm{d}x_n \otimes (\overline{f}_1 \wedge \ldots \wedge \overline{f}_n)^{\otimes -1}$$

by definition.

Let $Y \to X$ be a locally quasi-finite syntomic morphism of locally Noetherian schemes. By Remarks 0BVG and 0BVJ we have a coherent $\mathcal{O}_Y$-module $\omega_{Y/X}$ and a canonical global section

$$\tau_{Y/X} \in \Gamma(Y, \omega_{Y/X})$$

which affine locally recovers the pair $\omega_{B/A}, \tau_{B/A}$. By Lemma 2.9 the module $\omega_{Y/X}$ is invertible. Suppose given a commutative diagram of locally Noetherian schemes

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

whose vertical arrows are locally quasi-finite syntomic and which induces an isomorphism of Y' with an open of $X' \times_X Y$. Then there is a canonical base change map

$$b^* \omega_{Y/X} \longrightarrow \omega_{Y'/X'}$$

which is an isomorphism mapping $\tau_{Y/X}$ to $\tau_{Y'/X'}$. Namely, the base change map in the affine setting is (0BVB), it is an isomorphism by Lemma 0BVF, and it maps $\tau_{Y/X}$ to $\tau_{Y'/X'}$ by Lemma 0BT9 part (1).

2. Comparing sheaves of differential forms

The goal of this section is to construct for every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ maps

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for all $p \geq 0$ satisfying the following properties

(1) $c^0_{Y/X}(1) = \delta(NL_{Y/X})$ see Discriminants, Section 0FKB,
(2) $c^{q+p}_{Y/X}(\omega \wedge \eta) = \omega \wedge c^p_{Y/X}(\eta)$ for local sections ω of $f^*\Omega^q_{X/\mathbf{Z}}$ and η of $\Omega^p_{Y/\mathbf{Z}}$,
(3) the formation of $c^p_{Y/X}$ is compatible with restriction to opens and with base change (as formulated in Lemma 2.4).

Note that these conditions imply that the maps $c^p_{Y/X}$ are $\mathcal{O}_Y$-linear and that the composition with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$. We will in fact show that there exists a unique collection of operators $c^p_{Y/X}$ with properties (1), (2), (3). This result will be applied in Section 0FK6 to the construction of the trace map on the de Rham complexes if f is finite.

Lemma 2.1. *Let R be a ring and consider a commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K^0 & \longrightarrow & L^0 & \longrightarrow & M^0 & \longrightarrow & 0 \\ & & & & \uparrow{\scriptstyle \partial} & & \uparrow & & \\ & & & & L^{-1} & = & M^{-1} & & \end{array}$$

of R-modules with exact top row and M^0 and M^{-1} finite free of the same rank. For $p \geq 0$ there are canonical maps

$$c^p : \wedge^p(H^0(L^\bullet)) \longrightarrow \wedge^p(K^0) \otimes_R \det(M^\bullet)$$

with the following properties:

(1) $c^0(1) = \delta(M^\bullet)$, *and*
(2) $c^{q+p}(\omega \wedge \eta) = \omega \wedge c^p(\eta)$ *for* $\omega \in \wedge^q(K^0)$ *and* $\eta \in \wedge^p(H^0(L^\bullet))$.

In particular, the composition of c^p with the map $\wedge^p(K^0) \to \wedge^p(H^0(L^\bullet))$ is multiplication with $\delta(M^\bullet)$.

Proof. Say M^0 and M^{-1} are free of rank n. For every $p \geq 0$ there is a unique surjection

$$\pi_p : \wedge^{p+n}(L^0) \longrightarrow \wedge^p(K^0) \otimes \wedge^n(M^0)$$

with the following two properties; (i) we have $\pi_p(k_1 \wedge \ldots \wedge k_p \wedge l_1 \wedge \ldots \wedge l_n) = k_1 \wedge \ldots \wedge k_p \otimes m_1 \wedge \ldots \wedge m_n$ if $k_1, \ldots, k_p \in K^0$ and $l_1, \ldots, l_n \in L^0$ map to $m_1, \ldots, m_n \in M^0$ and (ii) the kernel of π_p is the submodule generated by wedges $l_1 \wedge \ldots \wedge l_{p+n}$ such that $> p$ of the l_j are in K^0. Pick a basis $e_1, \ldots, e_n$ for $L^{-1} = M^{-1}$. Then we define our map by the rule

$$\eta \mapsto \pi_p(\tilde{\eta} \wedge \partial(e_1) \wedge \ldots \wedge \partial(e_n)) \otimes (e_1 \wedge \ldots \wedge e_n)^{\otimes -1}$$

where $\eta \in \wedge^p(H^0(L^\bullet))$ and $\tilde{\eta} \in \wedge^p(L^0)$ is a lift of η. The expression on the right hand side makes sense because $\det(M^\bullet) = \wedge^n(M^0) \otimes \wedge^n(M^{-1})^{\otimes -1}$. The expression on the right is independent of the choice of $\tilde{\eta}$ since $\partial(\xi) \wedge \partial(e_1) \wedge \ldots \wedge \partial(e_n) = 0$ for any $\xi \in L^{-1}$. A straightforward calculation shows that the map is independent of the choice of the basis of $L^{-1} = M^{-1}$. It is immediate from the definition of

$\delta(M^\bullet)$ in More on Algebra, Section 0FJI that $c^0(1) = \delta(M^\bullet)$. Finally, the rule in statement (2) can be directly verified from the definition. □

Lemma 2.2. *Let $R_1 \to R_2$ be a ring homomorphism. For $i = 1, 2$ consider commutative diagrams*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K_i^0 & \longrightarrow & L_i^0 & \longrightarrow & M_i^0 & \longrightarrow & 0 \\ & & & & \uparrow{\scriptstyle \partial_i} & & \uparrow & & \\ & & & & L_i^{-1} & = & M_i^{-1} & & \end{array}$$

of R_i-modules as in Lemma 2.1. Assume we have maps $K_1^0 \to K_2^0$, $L_1^j \to L_2^j$, $M_1^j \to M_2^j$ compatible with the given ring map $R_1 \to R_2$ and compatible with the maps in the displayed diagrams. If the maps $M_1^j \otimes_{R_1} R_2 \to M_2^j$ are isomorphisms, then the diagrams

$$\begin{array}{ccc} \wedge^p(H^0(L_1^\bullet)) & \xrightarrow[c^p]{} & \wedge^p(K_1^0) \otimes_{R_1} \det(M_1^\bullet) \\ \downarrow & & \downarrow \\ \wedge^p(H^0(L_2^\bullet)) & \xrightarrow{c^p} & \wedge^p(K_2^0) \otimes_{R_2} \det(M_2^\bullet) \end{array}$$

commute.

Proof. This follows from the explicit description of the map given in the proof of Lemma 2.1. Note that we need the assumption that $M_1^j \otimes_{R_1} R_2 \to M_2^j$ are isomorphisms to see that the basis we use for M_1^{-1} maps to a basis for M_2^{-1} (and hence in checking the compatibility in both constructions we can use the "same" basis). □

Remark 2.3. Let A be a ring. Let $P = A[x_1, \ldots, x_n]$. Let $f_1, \ldots, f_n \in P$ and set $B = P/(f_1, \ldots, f_n)$. Assume $A \to B$ is quasi-finite. Then B is a relative global complete intersection over A (Algebra, Definition 00SP) and $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the classes $\overline{f}_i$ by Algebra, Lemma 00SV. Consider the following diagram

$$\begin{array}{ccccc} \Omega_{A/\mathbf{Z}} \otimes_A B & \longrightarrow & \Omega_{P/\mathbf{Z}} \otimes_P B & \longrightarrow & \Omega_{P/A} \otimes_P B \\ & & \uparrow & & \uparrow \\ & & (f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 & = & (f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \end{array}$$

The right column represents $NL_{B/A}$ in $D(B)$ hence has cohomology $\Omega_{B/A}$ in degree 0. The top row is the split short exact sequence $0 \to \Omega_{A/\mathbf{Z}} \otimes_A B \to \Omega_{P/\mathbf{Z}} \otimes_P B \to \Omega_{P/A} \otimes_P B \to 0$. The middle column has cohomology $\Omega_{B/\mathbf{Z}}$ in degree 0 by Algebra, Lemma 00RU. Thus by Lemma 2.1 we obtain canonical B-module maps

$$\Omega^p_{B/\mathbf{Z}} \longrightarrow \Omega^p_{A/\mathbf{Z}} \otimes_A \det(NL_{B/A})$$

for $p \geq 0$ satisfying properties (1) and (2) and in particular satisfy the property that the composition with $\Omega^p_{A/\mathbf{Z}} \to \Omega^p_{B/\mathbf{Z}}$ is multiplication by $\delta(NL_{B/A})$.

Lemma 2.4. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic and assume given maps $c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \to f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$ for $p \geq 0$ satisfying (1) and (2). Then there is at most one collection of maps $c^p_{Y'/X'} : \Omega^p_{Y'/\mathbf{Z}} \to (f')^*\Omega^p_{X'/\mathbf{Z}} \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'})$ for $p \geq 0$ satisfying (1) and (2) such that the diagrams

$$\begin{array}{ccccc} b^*\Omega^p_{Y/\mathbf{Z}} & \xrightarrow[b^*c^p_{Y/X}]{} & b^*(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})) & = & (f')^*a^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} b^*\det(NL_{Y/X})) \\ \downarrow & & & & \downarrow \\ \Omega^p_{Y'/\mathbf{Z}} & & \xrightarrow{c^p_{Y'/X'}} & & (f')^*\Omega^p_{X'/\mathbf{Z}} \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'}) \end{array}$$

commute for all $p \geq 0$. Here the vertical arrows use the maps $b^*\Omega^p_{Y/\mathbf{Z}} \to \Omega^p_{Y'/\mathbf{Z}}$ and $a^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{X'/\mathbf{Z}}$ of Section 07HX as well as the identification $b^*\det(NL_{Y/X}) = \det(NL_{Y'/X'})$ of Discriminants, Section 0FKB.

Proof. The map

$$(f')^*\Omega_{X'/\mathbf{Z}} \oplus b^*\Omega_{Y/\mathbf{Z}} \longrightarrow \Omega_{Y'/\mathbf{Z}}$$

is surjective because Y' is an open subscheme of the fibre product; the corresponding algebra statement is that $\Omega_{B\otimes_A C/\mathbf{Z}}$ is generated as a module by the images of $\Omega_{B/\mathbf{Z}}$ and $\Omega_{C/\mathbf{Z}}$. Thus for all p the map

$$\bigoplus\nolimits_{i=0,\ldots,p} (f')^*\Omega^i_{X'/\mathbf{Z}} \otimes b^*\Omega^{p-i}_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{Y'/\mathbf{Z}}$$

is surjective. Conditions (1) and (2) combined with the commutativity of the diagram in the lemma, implies that the map $c^p_{Y'/X'}$ is determined (but existence is not immediate). □

Lemma 2.5. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic. Then for every $y' \in Y'$ we can find opens $V' \subset Y'$, $V \subset Y$, $U' \subset X$, $U \subset X$ with $y' \in V'$, with $f'(V') \subset U'$, $b(V') \subset V$, $a(U') \subset U$, $f(V) \subset U$, and such that for $p \geq 0$ there are maps $c^p_{V/U}$ and $c^p_{V'/U'}$ satisfying (1) and (2) which are compatible with the diagram

$$\begin{array}{ccc} V' & \xrightarrow[b]{} & V \\ \downarrow & & \downarrow \\ U' & \longrightarrow & U \end{array}$$

in the sense explained in Lemma 2.4.

Proof. It is clear that we may replace Y' by $X' \times_X Y$ in order to prove this. Pick affine opens $V \subset Y$ and $U \subset X$ with V containing the image of y' as in Discriminants, Lemma 0BWE part (5). Pick an affine open $U' \subset X'$ containing the image of y' and mapping into U. After replacing X, Y, X', Y' by $U, V, U', U' \times_U V$ we may assume we are in the situation described in the next paragraph.

Assume that $X' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, and $Y' = \mathrm{Spec}(A' \otimes_A B)$ with $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ a relative global complete intersection over A. Then also $B' = A'[x_1, \ldots, x_n]/(f'_1, \ldots, f'_n)$ is a relative global complete intersection

over A' where f_i' is the image of f_i in $A'[x_1, \ldots, x_n]$, see Algebra, Lemma 00SS. The construction in Remark 2.3 provides us with the maps $c^p_{V/U}$ and $c^p_{V'/U'}$. These maps are compatible by Lemma 2.2 and the compatibility of the presentations of B and B' over A and A'. Some details omitted. □

Proof. This proof is very similar to the proof of Discriminants, Proposition 0FKD and we suggest the reader look at that proof first.

Let us reformulate the statement. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $f : Y \to X$ of schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The lemma means that for every object Y/X of $\mathcal{C}$ we have maps $c^p_{Y/X}$, $p \geq 0$ with properties (1) and (2), and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have that $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 2.4.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine open $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ such that there exists some maps

$$\Omega^p_{Y/\mathbf{Z}}|_V \longrightarrow \left(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})\right)|_V$$

with properties (1) and (2). This follows from picking affine opens as in Discriminants, Lemma 0BWE part (5) and then using the construction in Remark 2.3.

Note that the étale locus of f is exactly the set of points where $\delta(NL_{Y/X})$ does not vanish, see discussion in Discriminants, Section 0FKB. In particular $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ becomes an isomorphism after inverting $\delta(NL_{Y/X})$. We conclude that if $\Omega^p_{X/\mathbf{Z}}$ is finite locally free and the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero, then the local maps constructed in the previous paragraph are unique and automatically glue!

Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that

(1) $\Omega_{X/\mathbf{Z}}$ is locally free, and
(2) the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero.

By the remarks in the previous paragraph, we see that for any object Y/X of $\mathcal{C}_{nice}$ we have unique maps $c^p_{Y/X}$, $p \geq 0$ satisfying conditions (1) and (2). If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}_{nice}$, then the maps $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 2.4: namely, locally there do exist compatible maps by Lemma 2.5 and the uniqueness just mentioned shows these maps agree with $c^p_{Y/X}$ and $c^p_{Y'/X'}$. In other words, we have solved the problem on the full subcategory $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c^p_{Y/X}$ the solution we've just found.

In fact, more generally, suppose we have a morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ such that Y/X is an object of $\mathcal{C}_{nice}$. The same argument, but this time using the uniqueness in Lemma 2.4, tells us there are maps $c^p_{Y'/X'}$, $p \geq 0$ satisfying conditions (1) and (2) compatible with the already constructed maps $c^p_{Y/X}$. Let us call this the base change $(b/a)^* c^p_{Y/X}$.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** The two base changes $(b_i/a_i)^* c^p_{Y_i/X_i}$, $i = 1, 2$ are equal. We will first show that the claim implies the lemma and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Discriminants, Example 0FK8. Then $Y_{n,d}$ and $Y_{n,d}$ are irreducible schemes of finite type and smooth over $\mathbf{Z}$. Namely, $X_{n,d}$ is a spectrum of a polynomial ring over $\mathbf{Z}$ and $Y_{n,d}$ is an open subscheme of such. The morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Discriminants, Lemma 0FK9. Thus $\delta(NL_{Y_{n,d}/X_{n,d}})$ is nonzero: for example we have the local description of $\delta(NL_{Y/X})$ in Discriminants, Remark 0FKC and we have the local description of étale morphisms in Morphisms, Lemma 02GU part (8). Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Discriminants, Lemma 0FKA we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Then we have the base change $c^p_{V/U} = (b/a)^* c^p_{Y_{n,d}/X_{n,d}}$. The claim guarantees these locally constructed maps $c^p_{V/U}$ glue! Thus we get a well defined global maps $c^p_{Y/X}$ with properties (1) and (2). Finally, let $b/a : Y'/X' \to Y/X$ be an arbitray morphism of $\mathcal{C}$. We have the maps $c^p_{Y'/X'}$, $p \geq 0$ and $c^p_{Y/X}$, $p \geq 0$ constructed in this paragraph. To check they are compatible as in Lemma 2.4 we may work locally on Y'/X' and Y/X. Thus we may assume that there exists a morphism $Y/X \to Y_{n,d}/X_{n,d}$. By construction, both the family $c^p_{Y'/X'}$ and the family $c^p_{X/Y}$ are compatible with the map towards $Y_{n,d}/X_{n,d}$. It follows easily from this (and the uniqueness in Lemma 2.4) that $c^p_{Y'/X'}$ is compatible with $c^p_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma 0H9B. Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A'_1 = A_1[x_\alpha] \to A \quad \text{and} \quad A'_2 = A_2[y_\beta] \to A$$

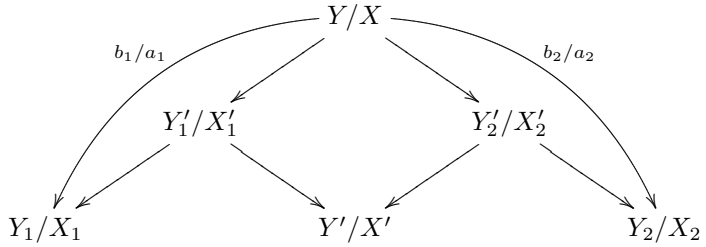
of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A'_i \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

$$\begin{array}{ccccc} (A'_1 \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A'_2 \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A'_1 & \longrightarrow & A & \longleftarrow & A'_2 \end{array}$$

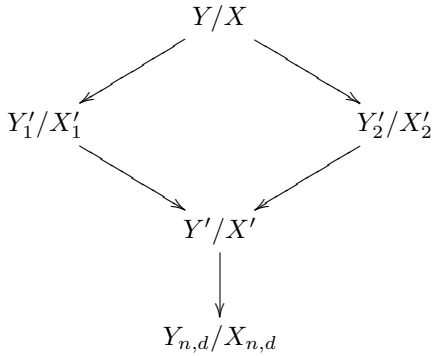
By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A'_1 \times_A A'_2 \longrightarrow B' = (A'_1 \otimes_{A_1} B_1)_{h_1} \times_B (A'_2 \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma 07RU we have $A'_1 \otimes_{A'} B' = (A'_1 \otimes_{A_1} B_1)_{h_1}$ and $A'_2 \otimes_{A'} B' = (A'_2 \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Discriminants, Lemma 0BWE). By More on Algebra, Lemma 08KQ the ring map $A' \to B'$ is flat and of finite presentation. Thus by Discriminants, Lemma 0BWE part (3) we conclude that Y'/X' is an object of $\mathcal{C}$. Consider now the commutative diagram



where Y'_i/X'_i corresponds to $A'_i \to (A'_i \otimes_{A_i} B_i)_{h_i}$. Note that Y'_i/X'_i is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an infinite dimensional affine space with Y_i/X_i; small detail omitted. In particular, the pullback of $c^p_{Y_i/X_i}$ via (b_i/a_i) is the same as the pullback of $c^p_{Y'_i/X'_i}$ via $Y/X \to Y'_i/X'_i$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$, but this is almost never the case. Namely, then pulling back $c^p_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X'_1, Y'_1, X'_2, Y'_2, X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c^p_{Y_{n,d}/X_{n,d}}$. This finishes the proof. □

Lemma 2.6. *Let $f : Y \to X$ be a morphism of schemes. The following are equivalent*

1. *f is locally quasi-finite and syntomic,*
2. *f is locally quasi-finite, flat, and a local complete intersection morphism,*
3. *f is locally quasi-finite, flat, locally of finite presentation, and the fibres of f are local complete intersections,*
4. *f is locally quasi-finite and for every $y \in Y$ there are affine opens $y \in V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ an integer n and $h, f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$ such that $B = A[x_1, \ldots, x_n, 1/h]/(f_1, \ldots, f_n)$,*

(5) *for every $y \in Y$ there are affine opens $y \in V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ such that $A \to B$ is a relative global complete intersection of the form $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$,*
(6) *f is locally quasi-finite, flat, locally of finite presentation, and $NL_{Y/X}$ has tor-amplitude in $[-1, 0]$, and*
(7) *f is flat, locally of finite presentation, $NL_{Y/X}$ is perfect of rank 0 with tor-amplitude in $[-1, 0]$,*

Proof. The equivalence of (1) and (2) is More on Morphisms, Lemma 069K. The equivalence of (1) and (3) is Morphisms, Lemma 01UF.

If $A \to B$ is as in (4), then $B = A[x, x_1, \ldots, x_n]/(xh - 1, f_1, \ldots, f_n)$ is a relative global complete intersection by see Algebra, Definition 00SP. Thus (4) implies (5). It is clear that (5) implies (4).

Condition (5) implies (1): by Algebra, Lemma 00SW a relative global complete intersection is syntomic and the definition of a relative global complete intersection guarantees that a relative global complete intersection on n variables with n equations is quasi-finite, see Algebra, Definition 00SP and Lemma 00PK.

Either Algebra, Lemma 00SY or Morphisms, Lemma 01UE shows that (1) implies (5).

More on Morphisms, Lemma 0FK3 shows that (6) is equivalent to (1). If the equivalent conditions (1) – (6) hold, then we see that affine locally $Y \to X$ is given by a relative global complete intersection $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ with the same number of variables as the number of equations. Using this presentation we see that

$$NL_{B/A} = \Big((f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} B\mathrm{d}x_i\Big)$$

By Algebra, Lemma 00SV the module $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the congruence classes of the elements $f_1, \ldots, f_n$. Thus $NL_{B/A}$ has rank 0 and so does $NL_{Y/X}$. In this way we see that (1) – (6) imply (7).

Finally, assume (7). By More on Morphisms, Lemma 0FK3 we see that f is syntomic. Thus on suitable affine opens f is given by a relative global complete intersection $A \to B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, see Morphisms, Lemma 01UE. Exactly as above we see that $NL_{B/A}$ is a perfect complex of rank $n - m$. Thus $n = m$ and we see that (5) holds. This finishes the proof. □

Lemma 2.7. *Invertibility of the relative dualizing module.*

(1) *If $A \to B$ is a quasi-finite flat homomorphism of Noetherian rings, then $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B \otimes_A \kappa(\mathfrak{p})/\kappa(\mathfrak{p})}$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for all primes $\mathfrak{p} \subset A$.*
(2) *If $Y \to X$ is a quasi-finite flat morphism of Noetherian schemes, then $\omega_{Y/X}$ is invertible if and only if $\omega_{Y_x/x}$ is invertible for all $x \in X$.*

Proof. Proof of (1). As $A \to B$ is flat, the module $\omega_{B/A}$ is A-flat, see Lemma 0BVE. Thus $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B/A} \otimes_A \kappa(\mathfrak{p})$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for every prime $\mathfrak{p} \subset A$, see More on Morphisms, Lemma 080Q. Still using that $A \to B$ is flat, we have that formation of $\omega_{B/A}$ commutes with base change, see Lemma 0BVF. Thus we see that invertibility of the relative dualizing module, in the presence of flatness, is equivalent to invertibility of the relative dualizing module for the maps $\kappa(\mathfrak{p}) \to B \otimes_A \kappa(\mathfrak{p})$.

Part (2) follows from (1) and the fact that affine locally the dualizing modules are given by their algebraic counterparts, see Remark 0BVG. □

Lemma 2.8. *Let k be a field. Let $B = k[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ be a global complete intersection over k of dimension 0. Then $\omega_{B/k}$ is invertible.*

Proof. By Noether normalization, see Algebra, Lemma 00OY we see that there exists a finite injection $k \to B$, i.e., $\dim_k(B) < \infty$. Hence $\omega_{B/k} = \mathrm{Hom}_k(B, k)$ as a B-module. By Dualizing Complexes, Lemma 0AX0 we see that $R\mathrm{Hom}(B, k)$ is a dualizing complex for B and by Dualizing Complexes, Lemma 0A71 we see that $R\mathrm{Hom}(B, k)$ is equal to $\omega_{B/k}$ placed in degree 0. Thus it suffices to show that B is Gorenstein (Dualizing Complexes, Lemma 0DW9). This is true by Dualizing Complexes, Lemma 0DWA. □

Lemma 2.9. *Let $f : Y \to X$ be a morphism of locally Noetherian schemes. If f satisfies the equivalent conditions of Lemma 2.6 then $\omega_{Y/X}$ is an invertible $\mathcal{O}_Y$-module.*

Proof. We may assume $A \to B$ is a relative global complete intersection of the form $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ and we have to show $\omega_{B/A}$ is invertible. This follows in combining Lemmas 2.7 and 2.8. □

Example 2.10. Let $n \geq 1$ and $d \geq 1$ be integers. Let T be the set of multi-indices $E = (e_1, \ldots, e_n)$ with $e_i \geq 0$ and $\sum e_i \leq d$. Consider the ring

$$A = \mathbf{Z}[a_{i,E}; 1 \leq i \leq n, E \in T]$$

In $A[x_1, \ldots, x_n]$ consider the elements $f_i = \sum_{E \in T} a_{i,E} x^E$ where $x^E = x_1^{e_1} \ldots x_n^{e_n}$ as is customary. Consider the A-algebra

$$B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$$

Denote $X_{n,d} = \mathrm{Spec}(A)$ and let $Y_{n,d} \subset \mathrm{Spec}(B)$ be the maximal open subscheme such that the restriction of the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A) = X_{n,d}$ is quasi-finite, see Algebra, Lemma 00QA.

Lemma 2.11. *With notation as in Example 2.10 the schemes $X_{n,d}$ and $Y_{n,d}$ are regular and irreducible, the morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite and syntomic, and there is a dense open subscheme $V \subset Y_{n,d}$ such that $Y_{n,d} \to X_{n,d}$ restricts to an étale morphism $V \to X_{n,d}$.*

Proof. The scheme $X_{n,d}$ is the spectrum of the polynomial ring A. Hence $X_{n,d}$ is regular and irreducible. Since we can write

$$f_i = a_{i,(0,\ldots,0)} + \sum\nolimits_{E \in T, E \neq (0,\ldots,0)} a_{i,E} x^E$$

we see that the ring B is isomorphic to the polynomial ring on $x_1, \ldots, x_n$ and the elements $a_{i,E}$ with $E \neq (0, \ldots, 0)$. Hence $\mathrm{Spec}(B)$ is an irreducible and regular scheme and so is the open $Y_{n,d}$. The morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite and syntomic by Lemma 2.6. To find V it suffices to find a single point where $Y_{n,d} \to X_{n,d}$ is étale (the locus of points where a morphism is étale is open by definition). Thus it suffices to find a point of $X_{n,d}$ where the fibre of $Y_{n,d} \to X_{n,d}$ is nonempty and étale, see Morphisms, Lemma 02GU. We choose the point corresponding to the ring map $\chi : A \to \mathbf{Q}$ sending f_i to $x_i^d - 1$. Then

$$B \otimes_{A,\chi} \mathbf{Q} = \mathbf{Q}[x_1, \ldots, x_n]/(x_1^d - 1, \ldots, x_n^d - 1)$$

which is a nonzero étale algebra over $\mathbf{Q}$. □

Lemma 2.12. *Let $f : Y \to X$ be a morphism of schemes. If f satisfies the equivalent conditions of Lemma 2.6 then for every $y \in Y$ there exist n, d and a*

commutative diagram

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & Y_{n,d} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & X_{n,d} \end{array}$$

where $U \subset X$ and $V \subset Y$ are open with $y \in V$, where $Y_{n,d} \to X_{n,d}$ is as in Example 2.10, and where the square on the right hand side is cartesian.

Proof. By Lemma 2.6 we can choose U and V affine so that $U = \mathrm{Spec}(R)$ and $V = \mathrm{Spec}(S)$ with $S = R[y_1, \ldots, y_n]/(g_1, \ldots, g_n)$. With notation as in Example 2.10 if we pick d large enough, then we can write each g_i as $g_i = \sum_{E \in T} g_{i,E} y^E$ with $g_{i,E} \in R$. Then the map $A \to R$ sending $a_{i,E}$ to $g_{i,E}$ and the map $B \to S$ sending $x_i \to y_i$ give a cocartesian diagram of rings

$$\begin{array}{ccc} S & \longleftarrow & B \\ \uparrow & & \uparrow \\ R & \longleftarrow & A \end{array}$$

which proves the lemma. □

References

[1] Emmanuelle Garel. An extension of the trace map. *J. Pure Appl. Algebra*, 32(3):301–313, 1984.